

contact

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GitHub: theofuhrmann

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languages

Spanish (native)

French (native)

Catalan (native)

English (C1, fluent)

programming languagesPython, JavaScript, C++,
Scala**machine learning**PyTorch (+ Lightning),
Sklearn, TensorFlow,
WandB**Audio ML & DSP**

librosa, Essentia, RAVE

back-endDjango, PostgreSQL,
REST APIs, Node.js**cloud & devops**AWS, GCP, Docker,
SkyPilot, Apache Airflow**computer vision**

OpenCV, Pillow

music

Piano, Ableton

profile

Machine Learning Engineer working on **voice AI** models that improve audio quality and downstream transcription accuracy. Previously interned at SoundCloud building **large-scale audio recommendation pipelines**, and worked at Silt on **computer vision** models for digital ID verification. I hold a master's in **Sound and Music Computing** from UPF and a bachelor's in **Computer Science** from UPC.

education**Master's degree in Sound and Music Computing**

@ Universitat Pompeu Fabra (UPF)

2024–2025

Barcelona

Bachelor's degree in Computer Science

@ Universitat Politècnica de Catalunya (UPC)

2018–2022

Barcelona

- Specialization in Computation (AI, algorithms, data science and logic)

experience**Machine Learning Engineer**

Oct 2025-Pres

@ ai-coustics - Audio enhancement using AI

Berlin

- Co-developed and shipped multiple versions of Quail Voice Focus, a lightweight speech enhancement neural network that isolates the primary speaker and removes background voices, running in real-time on CPU.
- Reduced WER by 40% across major STT providers on Dawn Chorus, our open-source background-speech benchmark released on Hugging Face.
- Designed data augmentation pipelines replicating real-world acoustic conditions to cover diverse production use cases; developed voice activity detection (VAD) models.

Machine Learning Engineer Internship

Jun-Aug 2025

@ SoundCloud - Music streaming platform for artists and listeners

Berlin

- Developed and deployed the "Liked By" homepage module, surfacing playlists of liked songs from followed users; contributed to increased user engagement. Set up A/B tests with Statsig.
- Built Airflow DAGs for "Liked By" and album recommendation modules, triggering BigQuery SQL pipelines and Dataflow jobs to populate BigTable; contributed to Scala-based API integration.

AI Specialist in Computer Vision

Oct 2022 – Oct 2024

@ Silt - Digital identity verification platform using AI and computer vision

Barcelona, Remote

- Engineered neural networks using convolutional layers and known backbones with PyTorch and monitored their performances with WandB.
- Developed ML models for multiple identity documents (national IDs, passports, driving licenses, etc.), selfie, and selfie video recognition.
- Generated multiple synthetics datasets (MRZ images, IDs and MRZs with organic overlays, etc.) for data augmentation.
- Improved the accuracy of previous anti-spoof models by developing an image recapture detector using the aforementioned datasets, while also expanding the spoof categories.
- Performed multiple back-end and MLOps tasks, such as integrating ML/LLM models with the back-end infrastructure and optimizing the company's ML lifecycle.

Full-Stack Developer

Feb-July 2022

@ Discoverfy - SaaS platform with interactive e-commerce tools

Barcelona

- Built front-end and back-end features for an interactive e-commerce SaaS, including Shopify integration, using React, Node.js, and PostgreSQL.

projects**Understanding Audio Source Separation in Carnatic Music with Multimodal Data**

Aug 2025

- Master's thesis on the exploration of audio source separation models with multimodal data.
- Combined computer vision, audio processing and ML interpretability techniques.

Weathered Chaos

Mar 2025

- Music conditioned by the current weather and double pendulums using RAVE and MIDI.

Liquiprism

Dec 2024

- Implemented the LiquiPrism paper and developed an immersive visualizer and sonifier.